

D.5 — Mass–Spring–Damper: Transfer-function model

Given the standard equation of motion (zero initial conditions):

$$m \ddot{z}(t) + b \dot{z}(t) + k z(t) = F(t).$$

(a) Laplace transform (time $\rightarrow$ s-domain).

$$m s^2 Z(s) + b s Z(s) + k Z(s) = F(s).$$

(b) Transfer function $G_{zF}(s) = Z(s)/F(s)$.

$$G_{zF}(s) = \frac{1}{ms^2 + bs + k} = \frac{\frac{1}{m}}{s^2 + \frac{b}{m}s + \frac{k}{m}}.$$

(c) Block diagram (text description).

- Single block $G_{zF}(s) = 1/(ms^2 + bs + k)$ from input F to output z .
- Equivalent signal flow:
 - Sum node (net force): $F - b \dot{z} - k z$.
 - Gain $1/m \rightarrow$ acceleration $\ddot{z}$.
 - Two cascaded integrators $1/s \rightarrow \dot{z}$ then z .
 - Feedback taps: k from z and b from $\dot{z}$ into the sum node.

D.6 — Mass–Spring–Damper: State-space model

Let

$$x = \begin{bmatrix} z \\ \dot{z} \end{bmatrix}, \quad u = F, \quad y = z.$$

From $m \ddot{z} + b \dot{z} + k z = F$:

\$\$

\dot{x}=

\begin{bmatrix}

$\dot{z}$ $\ddot{z}$

$\end$

```
\underbrace{\begin{bmatrix}
0 & 1[2pt]
-\tfrac{k}{m} & -\tfrac{b}{m}
\end{bmatrix}}{A}x
+
\underbrace{\begin{bmatrix}
0[2pt]\tfrac{1}{m}
\end{bmatrix}}{B}u,
\quad
y=\underbrace{\begin{bmatrix}1;0\end{bmatrix}}{C}x+\underbrace{\begin{bmatrix}0\end{bmatrix}}{D}u.
$$
```

HW3 Testing — D Section (Spring Mass)

1. [massDynamics.py](#)

Run [massDynamics.py](#)

Expect all Tests (1-10) to pass

C:/Users/consa/Downloads/Programming/Robotics_Controls/.venv/Scripts/python.exe c:/Users/consa/Downloads/Programming/

Output

```
Test 1: PASS
Test 2: PASS
Test 3: PASS
Test 4: PASS
Test 5: PASS
Test 6: PASS
Test 7: PASS
Test 8: PASS
Test 9: PASS
Test 10: PASS
```

Excellent work!! The $f(x,u)$ function from your dynamics file has passed all of the tests!

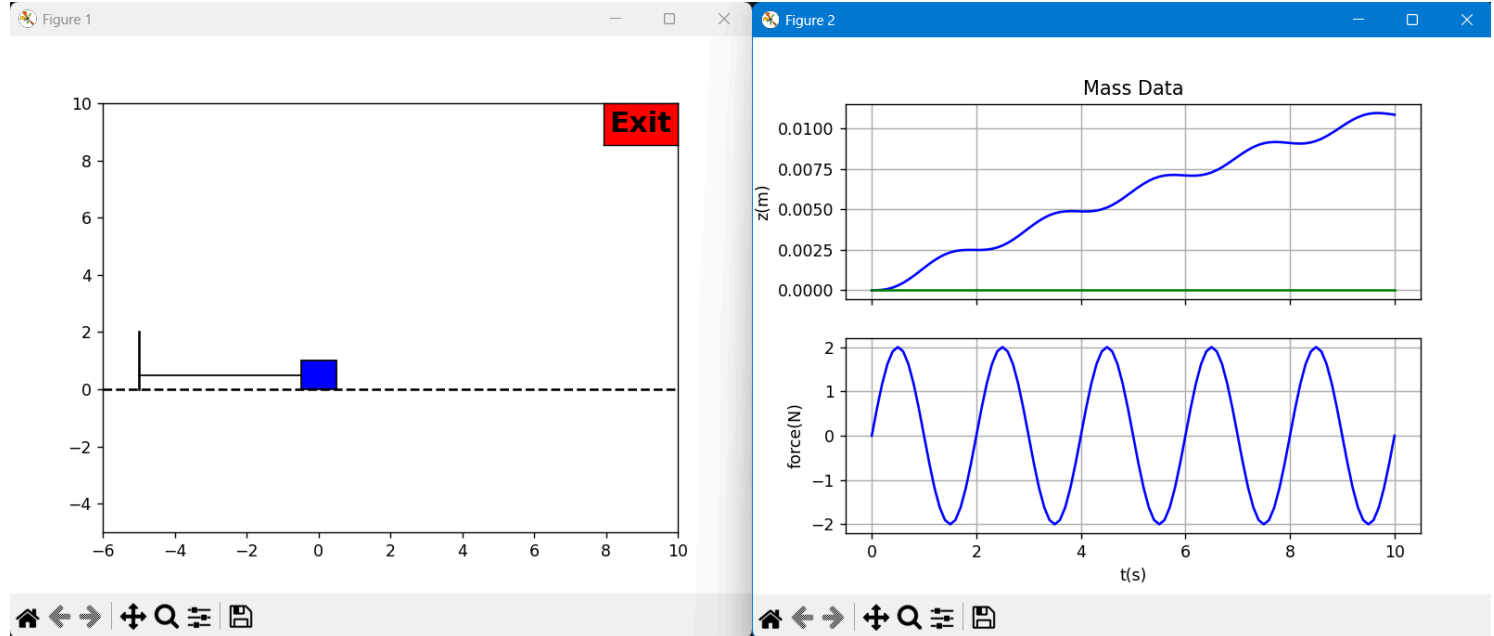
2. runMassAnimation.py

Run runMassAnimation.py

C:/Users/consa/Downloads/Programming/Robotics_Controls/.venv/Scripts/python.exe c:/Users/consa/Downloads/Programming/

Output

Expect smooth oscillatory position response to the sinusoidal force



D.5 transfer function Z/F: num = [1.] den = [5. 0.5 3.]

D.6 state space:

A=

```
[[ 0.  1. ]
 [-0.6 -0.1]]
```

B=

```
[[0. ]
 [0.2]]
```

C=

```
[[1. 0.]]
```

D=

```
[[0.]]
```

Close the figure window to end.